

Cameron CALK

Current employment

Postdoctoral researcher : Laboratoire d'Informatique et Systèmes (UMR 7020)

Institution : Aix-Marseille Université

Project : ANR [DisQC](#)

Previous employment and education

- 2022-2023 **Postdoctoral researcher**, *Université Aix-Marseille*, Laboratoire d'Informatique et des Systèmes (LIS) UMR 7020, Luminy, in the ANR [DUCAT](#) project.
with Emmanuel Godard
- 2022-2023 **Postdoctoral researcher**, *Université Aix-Marseille*, Laboratoire d'Informatique et des Systèmes (LIS) UMR 7020, Luminy, in the ANR [LambdaComb](#) project.
with Luigi Santocanale
- 2019-2022 **Doctorate**, *École Polytechnique*, Laboratoire d'Informatique de l'Ecole Polytechnique (LIX) UMR 7161, Paris-Saclay, "*Homotopy of directed calculation*" with jury's felicitations.
Advisor : Eric Goubault, Co-advisor : Philippe Malbos.
- 2017-2018 **Master in advanced mathematics**, *ENS de Lyon & Université Claude Bernard Lyon 1*
Geometry specialization, Cum Laude.
- 2016-2017 **Master 1 in advanced mathematics**, *ENS de Lyon & Université Claude Bernard Lyon 1*, Cum Laude
- 2015-2016 **Bachelors in advanced mathematics**, *ENS de Lyon & Université Claude Bernard Lyon 1*
- 2015-2019 **Normalien**, *ENS de Lyon*
2015 **Second concours**, *ENS de Lyon*, Admitted, Ranked 2nd out of 12.
- 2013-2015 **Preparation for Magistère and Concours in Physics (PMCP)**, 1st and 2nd years of bachelors, Paris-Sud, Orsay
- 2011 **A-levels**, *Advanced mathematics, Mathematics, Philosophy and Physics*, Magdalen College School, Oxford, England
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- 2010 **AS-level**, *Chemistry*, Magdalen College School, Oxford, Angleterre
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Prizes

- 2022 **Best Thesis Prize**, of the *departement Informatique, Data et Intelligence Artificielle (IDIA)*, at Institut Polytechnique de Paris (IPP), Runner-up

Publications

- 2025 ***Higher catoids, higher quantales and their correspondences***, with P. Malbos, D. Pous and G. Struth, *Applied Categorical Structures*, [Volume 33, Article 25](#)
- 2024 ***Complete congruences of completely distributive lattices***, with L. Santocanale, RAMICS 2024, LNCS proceedings, [Volume 14787](#)

- 2024 *Persistent homology of directed spaces*, with E. Goubault and P. Malbos, Journal of Applied and Computational Topology, [Volume 8, p. 2247-2284](#)
- 2022 *Algebraic coherent confluence and higher-dimensional globular Kleene algebras*, with E. Goubault, P. Malbos and G. Struth, Logical Methods in Computer Science, [Volume 18, Issue 4](#)
- 2021 *Abstract strategies and coherence*, with E. Goubault and P. Malbos, RAMICS 2021, LNCS proceedings, [Volume 13027](#)
- 2021 *lr-Multisemigroups, modal quantales and the origin of locality*, with U. Fahrenberg, C. Johansen, G. Struth and K. Ziemanski, RAMICS 2021, LNCS proceedings, [Volume 13027](#)
- 2020 *Time-reversal homotopical properties of concurrent systems*, with E. Goubault and P. Malbos, Homology, Homotopy and Applications, [Volume 22, Number 2](#)

Pre-publications

- 2025 *Stone duality proofs for colorless distributed computability theorems*, with E. Godard, [arXiv:2511.03609](#)
- 2020 *Beyond formulas-as-cographs : an extension of Boolean logic to arbitrary graphs*, with A. Das and T. Waring, [arXiv:2004.12941](#)

Diffusion

- 2021-2023 *Algebraic Rewriting Seminar*, Bi-weekly, Virtual, [Website](#)
Co-organizer

Talks

- 6 Oct. 2025 *Introduction to Category Theory for Algebraic Topology (II)*, Séminaire TopoCS, LIS, Marseille
- 29 Sept. 2025 *Introduction to Category Theory for Algebraic Topology (I)*, TopoCS seminar, LIS, Marseille
- 5 June 2024 *Description of distributive executions using partial orders*, Meeting DUCAT, Le Pleyne
- 15 mars 2024 *Persistence and the classification of directed spaces*, Séminaire APR, LIP6, Jussieu, Paris
- 1 Feb. 2024 *Binomial lattice quotients and flat dihomotopy types*, Séminaire LdP, I2M, Université Aix-Marseille, Marseille
- 25 Jan. 2024 *From rewriting to quantales, and on to directed topology*, Séminaire PPS, IRIF, Université Paris Cité, Paris
- 23 Jan. 2024 *Binomial lattice quotients and flat dihomotopy types*, LambdaComb Days, Université Paris Sorbonne, Paris
- 18 Jan. 2024 *From rewriting to quantales, and on to directed topology*, Séminaire LoVe, LIPN, Université Paris-Nord, Villetaneuse
- 15 Dec. 2023 *Lattices of paths and flat dihomotopy types*, Computational Logic and Applications, Jagiellonian University, Kraków
- 26 Nov. 2023 *Coherence via rewriting in higher Kleene algebras*, Demi-journée du Pole Calcul, LIS, Université Aix-Marseille, Luminy
- 5 Dec. 2022 *Coherence and strategic rewriting in higher Kleene algebras*, Séminaire LIRICA, LIS, Université Aix-Marseille, Luminy
- 25 Fev. 2022 *Abstract strategies and coherence*, Logic in Higher Structures, Centre international de rencontres mathématiques (CIRM), Marseille

- 3 Nov. 2021 ***Abstract strategies and coherence***, RAMICS '21, Centre international de rencontres mathématiques (CIRM), Marseille
- 18 Juin 2021 ***Abstract strategies and coherence***, GT ALGA, Université de Bordeaux, Virtual
- 18 Fev. 2021 ***Strategies and resolutions***, Algebraic Rewriting Seminar, Virtual
- 5 Fev. 2021 ***Geometric realisation of constructible polygraphs***, Journées LHC (GDR CNRS), Virtual
- 14 Jan. 2021 ***Abstract rewriting***, Algebraic Rewriting Seminar, Virtual
- 30 Juin 2020 ***Coherent confluence in modal n -Kleene algebras***, International Workshop on Confluences 2020, Paris
- 8 Mai 2020 ***Coherent confluence in modal n -Kleene algebras***, Séminaire Catégories à l'IRIF (Paris 7), Paris
- 18 Juin 2019 ***Time-reversal homotopical properties in concurrency***, Topology, Algebra and Categories in Logic (TACL'19), Université de Nice
- 2 Fev. 2019 ***Time-reversal homotopical properties of concurrent systems***, Algebra Seminar (NMSU), Las Cruces, N.M., U.S.A.
- 14 Dec. 2018 ***L'homotopie dirigée et l'inversion temporelle***, Séminaire Catégories à l'IRIF (Paris 7), Paris
- 18 Oct. 2018 ***Time-reversal homotopical properties of concurrent systems***, Journées inaugurales LHC (GDR CNRS), Marseille

Participation in conferences, schools and workshops

- 12-14 Nov. 2025 **LambdaComb days**, CIRM, Marseille
Participant
- 19-23 May 2025 **EPIT'25, (Co)induction and circular reasoning**, Aussois
Participant
- 3-7 June 2024 **DUCAT meeting**, Le Pleyhet
Speaker
- 22-24 Jan. 2024 **LambdaComb days**, Jussieu, Sorbonne Université, Paris
Speaker
- 14-15 Dec. 2023 **17th Computational Logic and Applications**, Jagiellonian University, Cracovie
Speaker
- 21-25 Fev. 2022 **Logic in Higher Structures**, Centre international de rencontres mathématiques (CIRM), Marseille
Speaker
- 2-5 Nov. 2021 **RAMICS '21**, Centre international de rencontres mathématiques (CIRM), Marseille
Speaker
- 16-18 Juin 2021 **GT ALGA**, Université de Bordeaux, Virtual
Speaker
- 5 Fev. 2021 **Journées LHC 2021**, Virtual
Speaker
- 30 Juin 2020 **International Workshop on Confluence**, Paris, Virtual
Speaker
- 12-16 Avril 2020 **Midlands Graduate School in the Foundations of Computing Science (MGS)**, University of Sheffield, Virtual
Participant

- 17 - 21 Juin **Topology, Algebra, and Categories in Logic (TACL)**, *Université de Nice*, Nice
2019 Speaker
- 10 - 15 Juin **TACL summer school**, *Ile de Porquerolles*
2019 Participant
- 17- 19 Oct. **Journées inaugurales LHC**, (*GDR CNRS*), Marseille
2018 Speaker
- 8 - 19 Jan. **Advanced school and workshop "Entropies and Soficity"**, *dans le cadre du M2*,
2017 Université Claude Bernard Lyon 1, Lyon
Participant
- 4 - 10 Juillet **Logic in Computer Science (LICS)**, *Columbia University*, New York City
2016 Student volunteer

Teaching

- 2021-2022 **CSE305 : Concurrent and Distributed Computing**, *Acitivité complémentaire d'enseignement (ACE)*, 12h, Bachelor program of École Polytechnique
- 2021-2022 **Chargé de TD INF361 : Introduction à Java**, *ACE*, 20h, École Polytechnique
- 2019-2022 **Chargé de TD MAA306 : Topics in Differential Geometry**, *ACE*, 54h, Bachelor program of École Polytechnique
- 2020 **Chargé de TD MAA206 : Quadratic Forms and Applications**, *ACE*, 16h, Bachelor program of École Polytechnique
- 2019-2021 **Chargé de TD CSE103 : Introduction to Algorithms**, *dans le cadre du ACE*, 64h, Bachelor program of École Polytechnique
- 2019-2021 **Tutorat MAT361**, *dans le cadre du ACE*, 8h, École Polytechnique
- 2017 **Observation and teaching**, *M1 internship*, 32h, Lycée Branly

Internships

- Feb - July **Research internship**, *Projet long de recherche (PLR)*, at Laboratoire d'informatique de l'École Polytechnique (LIX) UMR 7161, Paris Saclay, "*Homotopy of globular CW complexes and their realization as polygraphs*"
2019 Supervisor : Eric Goubault.
- Sept. - Dec. **Research internship**, *PLR*, at l'Institut Camille Jordan (ICJ) UMR 5208, Lyon, "*Higher-dimensional rewriting and Kleene algebras*"
2018 Supervisor : Philippe Malbos.
- March - July **Research internship**, *Master*, at Laboratoire d'informatique de l'École Polytechnique (LIX) UMR 7161, Paris Saclay, "*Directed homotopy and time-reversal*"
2018 Supervisor : Eric Goubault
- Mai - Juillet **Research internship**, *Bachelor*, at Laboratoire de l'Informatique du Parallelisme (LIP) UMR 5668, Lyon, "*The extension of Boolean logic to graphs*"
2016 Supervisors : Olivier Laurent et Anumpam Das

Languages

English, *Mother tongue*
Danish, *Mother tongue*
French, *Fluent*